METHOD 27 / A COMPLETE BEGINNER GUIDE

Cointegration and error correction

When do two FX log prices share a long-run anchor, and how does the short-run return pull toward it? Cointegration uses log price levels. Error correction adds the lagged spread to a return equation.

This is **not** return correlation. Correlated returns can still leave a non-mean-reverting spread. Teaching numbers only; experiments not started.

Read it in three parts

First: Spread in log space and FX percent. **Next:** Engle-Granger (1987) context, ECM with kappa. **Last:** Hand rows, beta is not lot size, limits.

Pages	What you will learn
2	Levels vs return correlation
3	FX 1.1000 to 1.1110 in logs
4	Spread e and parameters alpha, beta
5	Engle & Granger 1987
6-9	ECM formula and hand rows
10-11	Beta not hedge lots
12-14	Limits, checks, sources

Prepared for the folder: 27 - Cointegration and Error Correction
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02 / THE BASIC IDEA

What does cointegration ask?

```
e_t = ln(PY_t) - alpha - beta * ln(PX_t)
```

PY and PX are FX prices in consistent units. The spread e_t is a log-price gap. Cointegration means e_t is stationary (mean-reverting) even when each log price trends.

Error correction on returns

```
delta ln(PY_t) = c + kappa * e_{t-1} + lags + u_t
```

kappa negative often means last period's positive spread pulls today's ln(PY) return down, nudging e toward zero. Lags and u_t carry short-run dynamics.

Not Pearson on returns

Folder 01 correlation of returns does not test this level anchor. Two return series can correlate highly while e_t drifts.

FX percent math on one move

From **1.1000 to 1.1110** on PX:

```
Simple return = 0.01 (1.00% label)
ln(PX_new) - ln(PX_old) = ln(1.1110/1.1000) = about 0.009950
```

Cointegration works in log price levels ln(P_t). Returns enter the ECM as changes delta ln(P_t). Use natural logs consistently for both legs.

Percent labels describe returns; spread e_t lives in log-price units, not percent points.

04 / DATA RULES

Spread parameters

Estimate alpha and beta

Engle-Granger step one regresses ln(PY) on ln(PX) (or vice versa per your design) to obtain alpha hat and beta hat. Step two tests whether residuals e_t are stationary.

Same clock

Row t must be the same time for both prices. Gaps in quotes need a documented rule before v001 experiments.

FX quotes

State which currency is base and quote for PX and PY. Reversing a quote changes log levels and beta interpretation.

Research notes

Engle, R. F., & Granger, C. W. J. (1987) [1]

Econometrica, 55(2), 251-276. Two-step cointegration and error-correction representation linking long-run equilibrium and short-run adjustments.

Modern systems use Johansen and other multivariate tests; this guide stays with the beginner spread plus ECM form used in many FX pair notes.

Claims here

Teaching spread and one ECM row only. No claim that any major pair is cointegrated until you run tests on snapshot data.

06 / DICTIONARY

Meet the symbols

Symbol	Meaning
ln(P)	Natural log of price; requires P > 0.
alpha	Intercept in log-log relation.
beta	Slope on ln(PX); not automatic lot size.
e_t	Spread / equilibrium error at t.
kappa	Error-correction speed on e_{t-1}.
delta ln(PY)	First difference of log price Y.

Lags of delta ln(PY) and delta ln(PX) absorb short-run dynamics omitted in the tiny hand example.

07 / WHAT EACH PART DOES

The main equations

```
e_t = ln(PY_t) - alpha - beta * ln(PX_t)

delta ln(PY_t) = c + kappa * e_{t-1}
+ lag terms + u_t
```

Read kappa

If kappa = -0.35, a spread e_{t-1} = +0.10 adds -0.035 to today's delta ln(PY) before lags. Sign and magnitude depend on units of e.

Stationarity of e

Cointegration requires statistical evidence that e_t does not behave like a random walk. ADF or similar tests belong in your estimation notebook, not skipped silently.

Hand calc: spread rows

Teaching alpha = 0.050, beta = 1.100 (fixed for arithmetic):

t	PX	PY	ln(PX)	ln(PY)	e_t
1	1.1000	1.2500	0.09531	0.22314	0.06830
2	1.1110	1.2625	0.10526	0.23309	0.06731
3	1.1050	1.2580	0.09985	0.22952	0.06969

PX row 2 uses the familiar 1.1000 to 1.1110 move. e_t is not a return percent; compare it only across time with the same alpha and beta.

09 / WORKED EXAMPLE PART TWO

Hand calc: one ECM row

```
delta ln(PY)_2 = ln(PY_2) - ln(PY_1) = about 0.009950
(same magnitude as the log return on PY for that step)

e_1 = 0.068302
ECM part: c + kappa * e_1 = 0.001 + (-0.35) * e_1
= about -0.022906 (teaching c, kappa fixed)
```

Full estimation fits c, kappa, and lags by regression on a long sample. Here we show how one lagged spread enters one return row.

Correlated daily returns could still leave e_t drifting if the level anchor fails.

10 / TRADING CAUTION

Beta is not automatic lot size

Hedge ratio confusion

beta links log levels in a cointegrating regression. It is not automatically the FX lot ratio for PnL hedging without conversion through levels, volatility, and account currency.

Mean reversion is statistical

Even with cointegration, e_t can stay wide for long spans. Risk limits must use spread volatility, not only kappa sign.

Versus folder 01

High return correlation does not imply mean-reverting e_t. Report which object you estimated.

11 / WHAT THIS CANNOT PROVE

Limits

Not proof of profitability

Stationary spread models ignore costs, jumps, and regime breaks.

Structural breaks

Fixing alpha and beta across crisis periods can fail. Rolling or break tests may be needed in v001.

Multiple currencies

Triangular FX relationships may need multivariate cointegration rather than one pairwise spread.

Software sketch

```
from math import log
alpha, beta = 0.05, 1.1
px, py = 1.1110, 1.2625
lnpx, lnpy = log(px), log(py)
e = lnpy - alpha - beta*lnpx
kappa = -0.35
c = 0.001
print(c + kappa * e) # ECM contribution from e at t-1
```

Use statsmodels.tsa.stattools.coint for Engle-Granger tests and OLS for step one when you move beyond toy rows.

13 / RESEARCH TRAIL

Sources and further reading

[1] Engle, R. F., & Granger, C. W. J. (1987). Co-integration and error correction: representation, estimation, and testing. Econometrica, 55(2), 251-276.
Primary cointegration and ECM reference.
Open source

[2] Folder 01 Pearson Correlation.
Return correlation contrast.
Open source

Check your understanding

Do I use log prices or returns for e_t? Log prices inside e_t; returns enter the ECM as deltas. **Does high return correlation imply cointegration?** No. **Is beta my hedge lot?** Not automatically.

When experiments start, save alpha hat, beta hat, ADF p-value on e_t, and kappa with standard errors from the same snapshot window.

Workflow before experiments

1. Write the economic pair

Name PX and PY with quote directions and why a long-run anchor might exist (policy band, cross-rate identity, etc.).

2. Levels first, returns second

Build ln(P) series, estimate spread, test stationarity, then specify ECM lags.

3. Unit check

Verify 1.1000 to 1.1110 log increment matches your return column for PX.

4. Report all steps

Do not report only kappa without alpha, beta, and test outcomes.

Experiments not started. No live cointegration claims until v001 with frozen snapshot.